

Sec 1.8 – Extending the Idea of a Limit

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

We can consider one-sided limits:

When we write $\lim_{x \rightarrow 2} f(x)$, we are approaching the x -value of 2 from *both* sides.

If we write $\lim_{x \rightarrow 2^+} f(x)$, we are approaching the x -value of 2 only from *values greater than 2*, coming from the right side.

If we write $\lim_{x \rightarrow 2^-} f(x)$, we are approaching the x -value of 2 only from *values less than 2*, coming from the left side.

Definition

If $\lim_{x \rightarrow a} f(x)$ does not exist because $f(x)$ gets arbitrarily large on both sides of a , we write the $\lim_{x \rightarrow a} f(x) = \infty$. (Similarly with a limit equal to $-\infty$.) This can be described as “diverging to infinity.”

If either the one-sided limit or the two-sided limit as $x \rightarrow a$ is $\pm\infty$, then we say that there is a vertical asymptote at $x = a$.

Definition

If $f(x)$ gets as close to a number L as we please when x gets sufficiently large, then we write

$$\lim_{x \rightarrow \infty} f(x) = L.$$

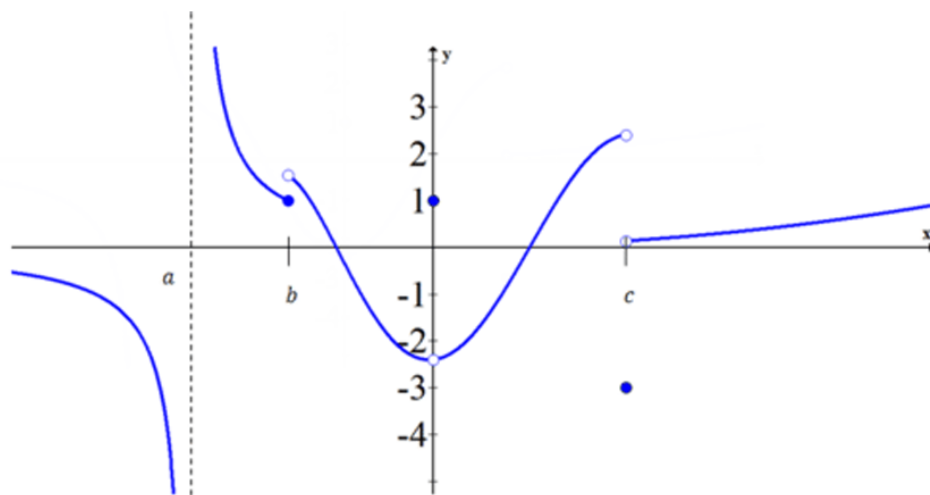
Similarly, if $f(x)$ approaches L when x is negative and has a sufficiently large absolute value, then we write

$$\lim_{x \rightarrow -\infty} f(x) = L.$$

In either case, we say that $y = L$ is a *horizontal asymptote* to the graph.

Partner up

Let $f(x)$ be as shown in the graph below.



Find the following limits:

- | | | | |
|---|-------------------------------------|--|-------------------------------------|
| (a) $\lim_{x \rightarrow -\infty} f(x)$ | (b) $\lim_{x \rightarrow a^-} f(x)$ | (c) $\lim_{x \rightarrow a^+} f(x)$ | (d) $\lim_{x \rightarrow a} f(x)$ |
| (e) $\lim_{x \rightarrow b^-} f(x)$ | (f) $\lim_{x \rightarrow b} f(x)$ | (g) $\lim_{x \rightarrow 0} f(x)$ | (h) $\lim_{x \rightarrow 0^+} f(x)$ |
| (i) $\lim_{x \rightarrow c^-} f(x)$ | (j) $\lim_{x \rightarrow c^+} f(x)$ | (k) $\lim_{x \rightarrow \infty} f(x)$ | |

Partner up

Note on the following, it is one function that satisfies all the properties.

Make ONE function that has all the following properties:

- ① $\lim_{x \rightarrow -\infty} f(x) = 2$
- ② $\lim_{x \rightarrow -3^-} f(x) = -2$
- ③ $\lim_{x \rightarrow -3^+} f(x) = 1$
- ④ $\lim_{x \rightarrow -2} f(x) = 3$, but $f(-2) = 1$
- ⑤ $\lim_{x \rightarrow 1} f(x) = \infty$
- ⑥ $\lim_{x \rightarrow 4^-} f(x) = -\infty$
- ⑦ $\lim_{x \rightarrow 4^+} f(x)$ does not exist
- ⑧ $\lim_{x \rightarrow \infty} f(x) = -4$

Theorem (1.2 Properties of Limits)

Suppose $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ both exist. Then the following are true:

- 1 For any constant k , $\lim_{x \rightarrow c} k = k$
- 2 $\lim_{x \rightarrow c} x = c$
- 3 **Constant Multiple Law** For any number b ,
 $\lim_{x \rightarrow c} (bf(x)) = b \left(\lim_{x \rightarrow c} f(x) \right)$.
- 4 **Sum Law** $\lim_{x \rightarrow c} (f(x) + g(x)) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x)$.
- 5 **Product Law** $\lim_{x \rightarrow c} (f(x)g(x)) = \left(\lim_{x \rightarrow c} f(x) \right) \left(\lim_{x \rightarrow c} g(x) \right)$.
- 6 **Quotient Law** $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$, as long as $\lim_{x \rightarrow c} g(x) \neq 0$.

Theorem (1.3 Continuity of Sums, Products, and Quotients of Functions)

If f and g are continuous on an interval and b is a constant, then, on that same interval,

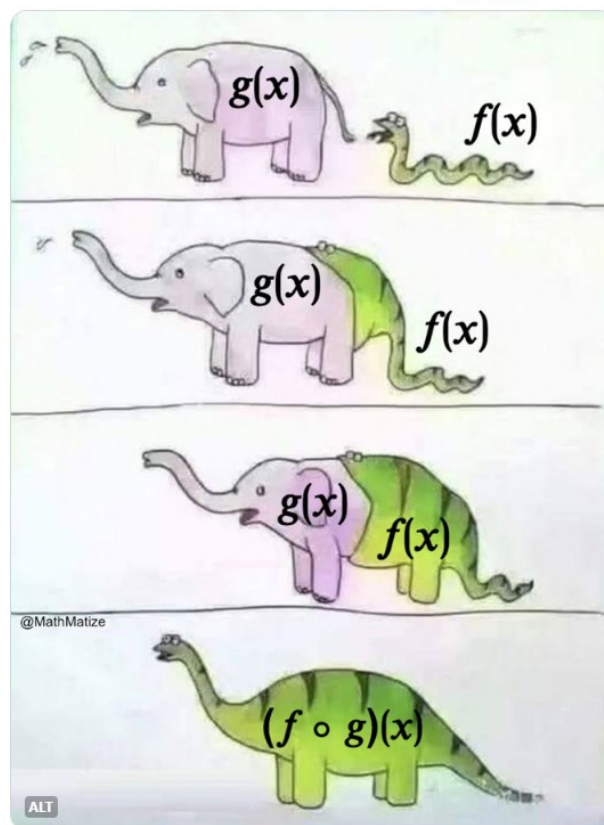
- 1 $bf(x)$ is continuous.
- 2 $f(x) + g(x)$ is continuous.
- 3 $f(x)g(x)$ is continuous.
- 4 $f(x)/g(x)$ is continuous, provided $g(x) \neq 0$ on the interval.

Theorem (1.4 Continuity of Composite and Inverse Functions)

If f and g are continuous, then

- 1 if the composite function $f(g(x))$ is defined on an interval, then $f(g(x))$ is continuous on that interval.
- 2 if f has an inverse function f^{-1} , then f^{-1} is continuous.

Dinosaurs are a composition of snakes and elephants 🐘🐍



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Use Theorem 1.3 to evaluate the limit:

$$\lim_{x \rightarrow 1} \frac{x^2 + 3 \sin x}{e^{x^2}}$$

We know that the functions x^2 , $\sin x$ and e^x are continuous since they are on our list of continuous functions.

By Theorem 1.3 (1), we know that $3 \sin x$ is continuous.

By Theorem 1.3 (2), we know that $x^2 + 3 \sin x$ is continuous.

By Theorem 1.4 (1), we know e^{x^2} is continuous.

By Theorem 1.3 (4), we know $\frac{x^2 + 3 \sin x}{e^{x^2}}$ is continuous anywhere

$e^{x^2} \neq 0$, which never happens.

Since the whole function is continuous, we can evaluate the limit simply by substituting 1:

$$\lim_{x \rightarrow 1} \frac{x^2 + 3 \sin x}{e^{x^2}} = \frac{1^2 + 3 \sin 1}{e^{1^2}} = \frac{1 + 3 \sin 1}{e}$$